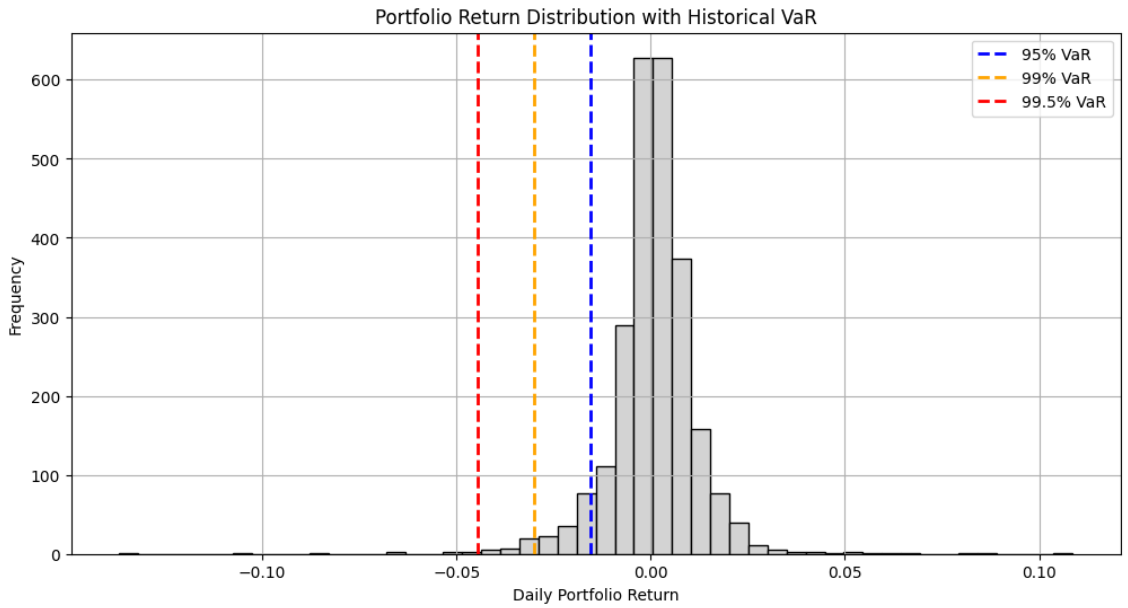
PORTFOLIO RISK RESEARCH / 2016-2025

Portfolio Market Risk & Tail-Loss Analysis

Value at Risk | Expected Shortfall | Volatility Models | Backtesting



Nivedhya Nair

Academic project report | Equal-weighted 50-stock US equity portfolio

50	2,513	1,763	4
equities	daily returns	holdout days	risk models

HISTORICAL SIMULATION | EWMA | GARCH | GJR-GARCH

01 / RESEARCH OVERVIEW

Abstract and study design

Abstract. This report evaluates daily downside risk in an equally weighted portfolio of 50 US equities using four approaches: rolling Historical Simulation (HS), EWMA volatility scaling, GARCH(1,1), and GJR-GARCH(1,1,1). The sample contains 2,513 portfolio returns from 5 January 2016 to 31 December 2025. Model performance is assessed on a chronological holdout of 1,763 observations from 27 December 2018 to 31 December 2025, following a 750-observation estimation period. At the 99% level, GJR-GARCH records 11 exceptions and passes the Kupiec, Christoffersen independence, and conditional-coverage tests. This is evidence about this sample and specification, not a general claim of model superiority.

Research questions

Question	Empirical focus
How does volatility scaling change tail estimates?	Compare HS with EWMA, GARCH and GJR-GARCH forecasts.
Do model forecasts match realized exceptions?	Test unconditional coverage, exception independence and joint conditional coverage.
What risks are missed by a single VaR number?	Compare Expected Shortfall and historical/hypothetical stress scenarios.

Main results at 99% confidence

Finding	Interpretation
GJR-GARCH: 11 exceptions; all 3 tests pass	Closest model to nominal 1% exception frequency among the tested methods, by the notebook's ranking rule.
GARCH: 10 exceptions; 2 of 3 tests pass	Independence and conditional coverage pass; Kupiec p = 0.0468 is just below 0.05.
EWMA: 7 exceptions; 0 of 3 tests pass	Fewer breaches alone do not imply calibration; the observed rate is well below the 1% target.
Historical: 32 exceptions; 0 of 3 tests pass	The 1.82% exception rate exceeds the nominal 1% rate and independence is rejected.

Scope and reporting convention

The analysis follows the notebook's 2016-2025 sample and model outputs. VaR thresholds are shown as negative return cutoffs; a more negative value denotes a more severe loss threshold. Model conclusions are conditional on this portfolio, sample and estimation design.

Analytical sequence

Stage	Purpose	Main output
1. Data	Align adjusted prices and returns	Clean return panel
2. Portfolio	Apply equal constituent weights	Daily portfolio return series
3. Risk models	Estimate empirical and conditional tails	1-day VaR and ES
4. Evaluation	Compare forecasts with realized returns	Coverage tests and exception record
5. Scenarios	Replay crises and impose shocks	Cumulative loss and stressed tails

02 / SAMPLE AND MEASUREMENT

Data and portfolio construction

The notebook uses daily adjusted closing prices for a fixed 50-stock US equity universe and builds a daily equal-weighted portfolio. Prices are aligned by trading date; the return series begins one trading day after the first price observation. The primary analysis spans January 2016 through December 2025. SPY is used as a benchmark in the notebook and dashboard.

$$r(i,t) = \ln[P(i,t) / P(i,t-1)] \text{ and } r(p,t) = \Sigma(i=1\ldots N)\, w(i) \cdot r(i,t)$$

Daily log return for asset i; portfolio return is the weighted sum of constituent log returns. N = 50 and w(i) = 1/50 in the baseline portfolio.

Sample map

Component	Notebook setting
Asset universe	50 US equities; equally weighted at 2% each
Price observations	2,514 dates, 4 Jan 2016-31 Dec 2025
Portfolio return observations	2,513 daily log returns, 5 Jan 2016-31 Dec 2025
Initial estimation sample	750 observations, 5 Jan 2016-26 Dec 2018
Out-of-sample evaluation	1,763 observations, 27 Dec 2018-31 Dec 2025
Baseline notional used for dollar views	$1,000,000

Portfolio construction note

The notebook forms portfolio log returns by weighting constituent log returns. This is a transparent static-weight construction. It does not represent a daily rebalanced, transaction-cost-adjusted portfolio unless the stated weight convention is explicitly imposed at each observation. Dollar illustrations rebase the return series to a stated starting notional; they are not a forecast of account value.

Observed portfolio profile

Statistic	Notebook series
Annualized volatility	18.70%
Compounded cumulative return	+422.52%
Maximum drawdown	-36.14%
Worst daily return	-13.69%

These descriptive figures use the notebook's weighted log-return series, with cumulative wealth formed by exponentiating cumulative log returns. They describe the sample path and are not risk forecasts.

Benchmark and data limits

The historical crisis dataset is extended for the 2008 scenario. The notebook reconstructs that period from 46 constituents with available prices; other historical scenarios use 50. This makes the 2008 portfolio composition slightly different from the main-sample portfolio.

03 / VAR AND EXPECTED SHORTFALL

Risk measures and empirical design

All principal VaR estimates are one-day lower-tail return thresholds. For confidence level c, the tail probability is p = 1 - c. Historical Simulation uses the empirical quantile of a trailing return window; ES averages returns in the tail at or below that threshold.

$$VaR(c,t) = Q(p \mid F(t-1)) \quad ES(c,t) = E[r(t) \mid r(t) \le Q(p \mid F(t-1))]$$

Return-threshold convention used in the backtest. Loss magnitudes can be expressed as absolute values for dollar reporting.

Historical Simulation windows

Window	Tail observations at 99%	Trade-off
250 trading days	~2.5	More responsive to recent regimes; limited tail sample.
500 trading days	~5.0	Intermediate smoothing and historical breadth.
750 trading days	~7.5	More tail observations, but slower response to structural change.

Tail sample size by confidence level

The expected number of observations beyond an empirical quantile falls quickly as the confidence level rises. This is the core finite-sample constraint in non-parametric tail estimation:

Window	95% tail	99% tail	99.5% tail
250 days	12.5	2.5	1.25
500 days	25.0	5.0	2.5
750 days	37.5	7.5	3.75

A longer window increases tail support but carries older observations forward. A shorter window is more responsive when a new shock enters and leaves the estimation sample. This trade-off is why no single historical window is uniformly best across stable and stressed regimes.

Rolling calculation

At forecast date t	Historical Simulation operation
1	Take the preceding W portfolio returns through t-1.
2	Estimate the lower-tail quantile and average the returns beyond it.
3	Apply the resulting VaR / ES to date t, then advance the window by one observation.

Window exit and the ghosting effect

A large shock can keep a rolling historical quantile elevated while it remains in the window, then disappear abruptly when it rolls out. The resulting VaR step is a mechanical effect of window membership rather than a new market signal.

Interpret ES with VaR

VaR identifies a threshold. ES summarizes tail severity beyond that threshold. Because the notebook's main comparison reports negative return values, more negative VaR and ES correspond to larger downside estimates.

04 / EXPONENTIALLY WEIGHTED VARIANCE

EWMA volatility scaling

EWMA updates conditional variance using the previous variance and the lagged squared return. The daily decay factor is λ = 0.94, the RiskMetrics convention used in the notebook.

$$\sigma^2(t) = \lambda\,\sigma^2(t-1) + (1-\lambda)\,r^2(t-1),\ \lambda = 0.94$$

Recent shocks receive more weight; the fixed decay factor determines the speed of volatility response.

Filtered historical scaling

The volatility-scaled approach first standardizes historical returns by their contemporaneous EWMA volatility. It then scales those standardized shocks by the current forecast volatility. In compact form, a past shock is transported from its original regime to the present volatility regime. This dynamic scaling can respond to volatility clusters without replacing the empirical distribution of standardized shocks with a Gaussian tail.

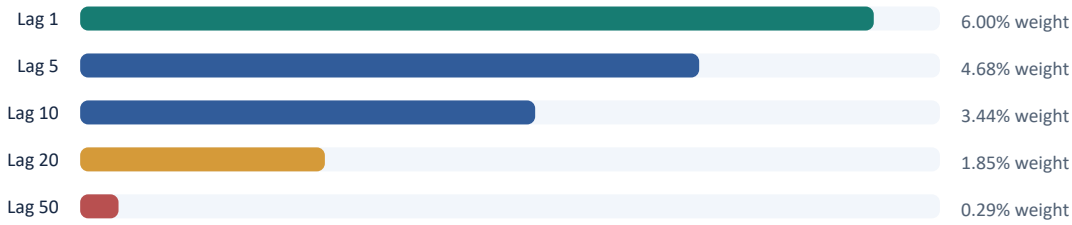
$$z(i) = r(i) / \sigma(i) \text{ and } r^*(i,t) = z(i) \cdot \sigma(t)$$

Historical standardized shocks z(i) are rescaled to the current conditional volatility σ(t) before the empirical tail is estimated.

Effect of the decay factor

Setting	Effect on memory	Implication
Higher λ	Slower decay of past variance	Smoother path; slower adjustment to a new shock
Lower λ	Greater weight on recent squared returns	Faster response; more day-to-day variation
Notebook λ = 0.94	Daily RiskMetrics convention	One fixed setting for reported results

Geometric decay of shock weights



Phase 5 full-sample diagnostic

Output	Notebook value
EWMA volatility range	0.19%-5.79% per day
EWMA-scaled 95% VaR	-0.9157%
EWMA-scaled 99% VaR	-1.7018%
EWMA-scaled 99.5% VaR	-2.2255%

These Phase 5 full-sample diagnostic thresholds are distinct from the 750-day estimation / 1,763-day evaluation results summarized later. They are included to show the model-development output and should not be interpreted as holdout performance.

EWMA diagnostic

EWMA records 7 breaches at 99% across 1,763 days, against 17.63 expected. A low breach count can reflect conservative thresholds rather than accurate calibration; coverage tests and tail-severity measures are needed alongside it.

05 / VOLATILITY CLUSTERING AND PERSISTENCE

GARCH(1,1) volatility model

GARCH(1,1) represents conditional variance as a baseline level, the previous innovation's squared magnitude, and the previous conditional variance. The specification captures volatility clustering: large shocks can raise near-term risk, while a high lagged variance sustains that risk after the initial move.

$$\sigma^2(t) = \omega + \alpha\, \varepsilon^2(t-1) + \beta\, \sigma^2(t-1)$$

GARCH(1,1) conditional variance recursion. The notebook uses normally distributed innovations.

Parameter interpretation

Term	Role in the recursion	Notebook full-sample estimate
ω	Positive variance baseline	0.038749
α	Immediate response to a new squared shock	0.172195
β	Persistence of past conditional variance	0.797457
α + β	Variance persistence summary	0.969652

These estimates describe the notebook's full-sample model-development fit. The formal backtest separately estimates the model on the first 750 observations, then uses recursive one-step-ahead variance forecasts for the remaining 1,763 days. This table is not the out-of-sample fit.

Persistence and shock half-life

$$\alpha + \beta = 0.969652 \quad h(1/2) = \ln(0.5) / \ln(\alpha + \beta) \approx 22.5 \text{ trading days}$$

Approximate decay half-life for a variance shock under the full-sample symmetric GARCH fit.

Because α + β is below one, the fitted symmetric variance recursion is stationary under the standard GARCH condition. Its value close to one indicates persistent volatility: a shock's contribution decays gradually rather than disappearing after a few sessions.

Risk scaling

For the out-of-sample forecast, the initial conditional variance comes from the end of the estimation period. After each realized return is observed, the recursion updates the variance input for the next forecast. VaR is obtained by applying the estimated standardized-return tail cutoff to forecast volatility. The notebook's alignment checks confirm that the first evaluation-date forecast follows the last estimation date.

What persistence means

A persistence sum near one implies that conditional variance shocks fade gradually. It does not imply that volatility is constant or that the model can anticipate an unprecedented shock.

06 / LEVERAGE RESPONSE

GJR-GARCH and asymmetric shocks

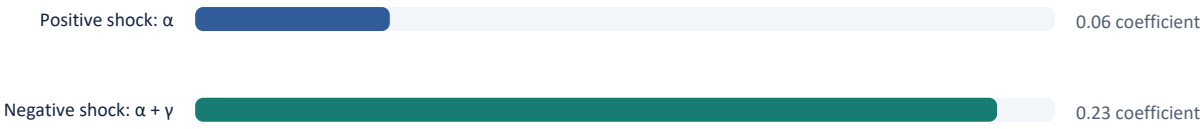
GJR-GARCH extends the symmetric GARCH recursion with an indicator that turns on for negative innovations. The added term allows a negative return shock to increase future conditional variance more than a positive shock with the same absolute size.

$$\sigma^2(t) = \omega + [\alpha + \gamma\, I(\varepsilon(t-1) < 0)]\, \varepsilon^2(t-1) + \beta\, \sigma^2(t-1)$$

GJR-GARCH(1,1,1); γ measures the incremental variance response associated with negative shocks.

In the full-sample fit, the notebook estimates γ = 0.1766 ($p < 0.001$). The positive estimate is statistically distinguishable from zero under the fitted model, consistent with an asymmetric response in this portfolio return sample.

Illustrative shock contribution using fitted full-sample coefficients



For this illustration, α = 0.0566 and γ = 0.1766 in the full-sample GJR-GARCH fit. Thus, the shock-driven coefficient is larger for a negative innovation. The chart compares the coefficient multiplying ε^2; realized variance impact also depends on shock magnitude and the remaining model terms.

Estimation and evaluation

Stage	Notebook treatment
Model-development fit	Full 2,513-return sample; descriptive parameters and volatility diagnostics.
Initial backtest fit	First 750 returns only; parameters are held fixed for recursive forecasts.
Forecast update	One-step-ahead conditional variance updates as each OOS return becomes known.
Performance evaluation	Forecast cutoff is compared with the aligned realized return.

Interpretation

Asymmetry is a model feature, not proof that all equity portfolios have the same leverage response. The backtest supports this specification on the notebook's holdout period; independent samples are needed to assess stability.

F1 / NOTEBOOK EXHIBIT

Volatility regimes across model families

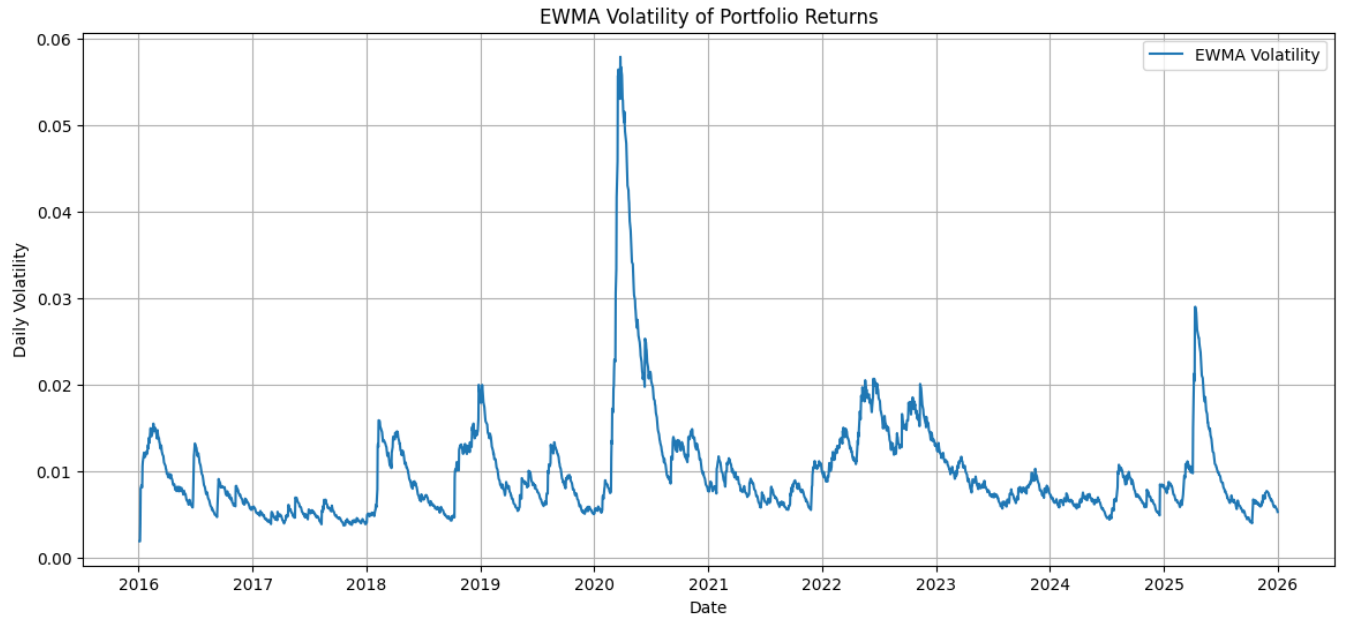


Exhibit 1a. EWMA conditional volatility using the notebook's fixed decay factor of 0.94.

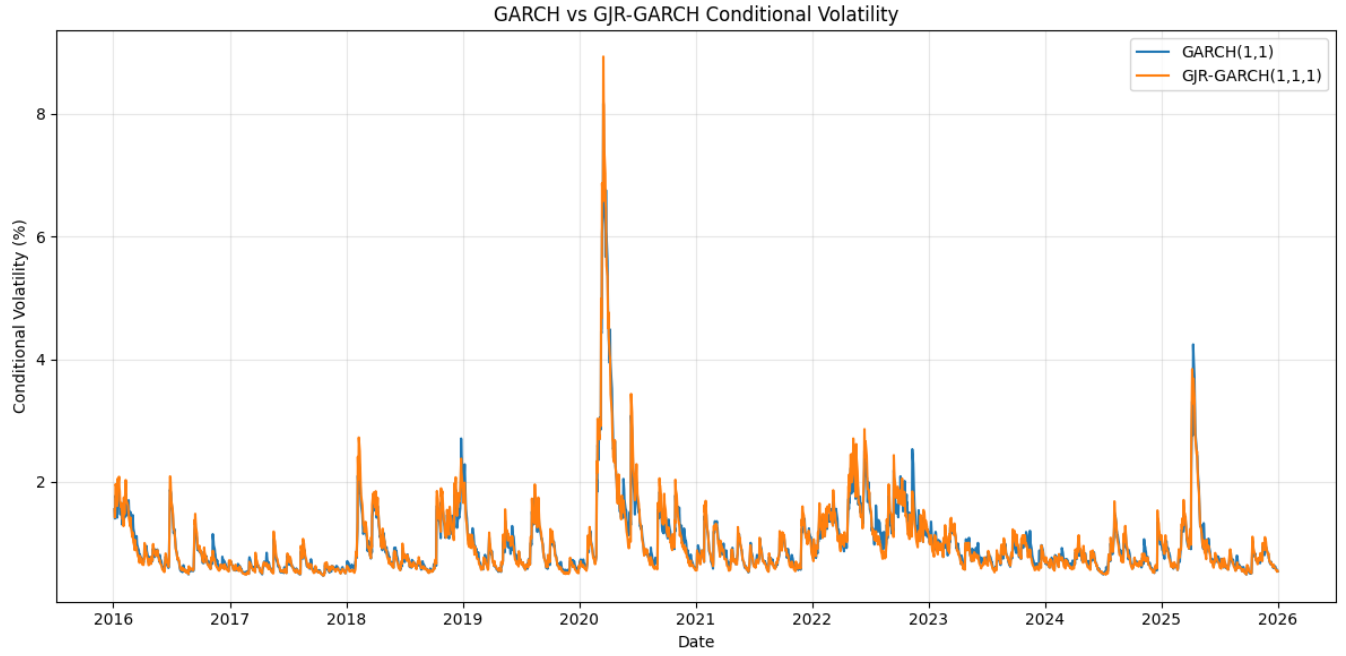


Exhibit 1b. GARCH(1,1) and GJR-GARCH(1,1,1) conditional volatility comparison.

Reading the chart

All three filters respond to volatility clustering, with the 2020 shock dominating the scale. The paired views show EWMA's smoother decay alongside the GARCH-family response and the additional asymmetry allowed by GJR-GARCH.

07 / FORECAST EVALUATION

Backtest design and test statistics

The backtest separates estimation from evaluation. The first 750 observations initialize model estimates; the subsequent 1,763 observations are evaluated sequentially. Each forecast is dated t and formed using information available before the realized return at t. Historical VaR is shifted one day; conditional-volatility recursions update after each observation.

Chronological split

Estimation period	Forecast period
5 Jan 2016 - 26 Dec 2018	27 Dec 2018 - 31 Dec 2025
750 returns	1,763 one-day evaluation dates

A return on the threshold boundary is not counted as an exception in the notebook. The lower-tail exception indicator is:

$$I(t) = 1\{ r(p,t) < VaR(c,t) \}$$

An exception is recorded when the realized portfolio return is below the forecast return cutoff.

Coverage diagnostics

Test	Null hypothesis	Question addressed
Kupiec POF	Observed exception probability = target tail probability	Is the overall breach rate consistent with the confidence level?
Christoffersen independence	Exception indicators are independent over time	Do exceptions cluster after previous exceptions?
Conditional coverage	Correct frequency and independence jointly	Does the model satisfy both coverage properties?

The notebook uses a 5% significance level. A p-value below 0.05 rejects the corresponding null. Failure to reject is not proof of calibration: finite samples can have limited power, especially for rare tail events.

Basel traffic light is reported as a separate diagnostic over the final 250 out-of-sample trading days at 99% VaR. It does not enter the notebook's model ranking, which uses the three statistical tests and distance from the 1% exception-rate target.

Forecast integrity

The notebook reports that the first out-of-sample date follows the last estimation date, all 1,763 return dates match the VaR forecast dates, and the recursive forecast consistency audit passes for all four models.

08 / COVERAGE AND EXCEPTION TIMING

Out-of-sample backtesting results

A VaR exception occurs when the realized return is below the forecast lower-tail cutoff. The notebook evaluates 1,763 out-of-sample days. Expected exceptions are 88.15 at 95%, 17.63 at 99%, and 8.82 at 99.5%. A correctly calibrated model should produce an exception frequency near the nominal tail probability over a sufficiently informative sample.

Exception counts across confidence levels

Model	95% (expected 88.15)	99% (expected 17.63)	99.5% (expected 8.82)
Historical	86	32	26
EWMA	103	7	1
GARCH	117	10	8
GJR-GARCH	124	11	9

99% tests (null not rejected when p ≥ 0.05)

Model	Exceptions / rate	Kupiec POF p	Independence p	Conditional coverage p	Tests passed
Historical	32 / 1.82%	0.0020	0.0022	0.0001	0 / 3
EWMA	7 / 0.40%	0.0038	0.0190	0.0010	0 / 3
GARCH	10 / 0.57%	0.0468	0.7355	0.1309	2 / 3
GJR-GARCH	11 / 0.62%	0.0882	0.7101	0.2181	3 / 3

Kupiec POF tests unconditional exception frequency. Christoffersen independence tests first-order clustering; conditional coverage combines frequency and independence. The GJR-GARCH result is the only 99% model that does not reject any of these three null hypotheses at 5%. GARCH's Kupiec p-value is close to, but below, the threshold.

Basel traffic light: separate 250-day diagnostic

Final window	Historical	EWMA	GARCH	GJR-GARCH
2 Jan-31 Dec 2025; 99% VaR	4 · Green	3 · Green	2 · Green	2 · Green

The Green classifications apply only to the final 250-day window and do not overturn the full-sample test results. In particular, Historical Simulation is Green in this short window while failing all three 99% full-sample tests.

Number of tests passed across confidence levels

Confidence	Historical	EWMA	GARCH	GJR-GARCH
95%	1 / 3	3 / 3	1 / 3	1 / 3
99%	0 / 3	0 / 3	2 / 3	3 / 3
99.5%	0 / 3	1 / 3	3 / 3	3 / 3

The model ranking in the notebook is deliberately based on the 99% comparison, while this matrix shows that the pattern varies by confidence level. At 95%, EWMA passes all three tests; at 99.5%, GARCH and GJR-GARCH pass all three. This variation argues against treating a single confidence level as a universal model verdict.

F2 / NOTEBOOK EXHIBIT

Backtest evidence: VaR paths and exceptions

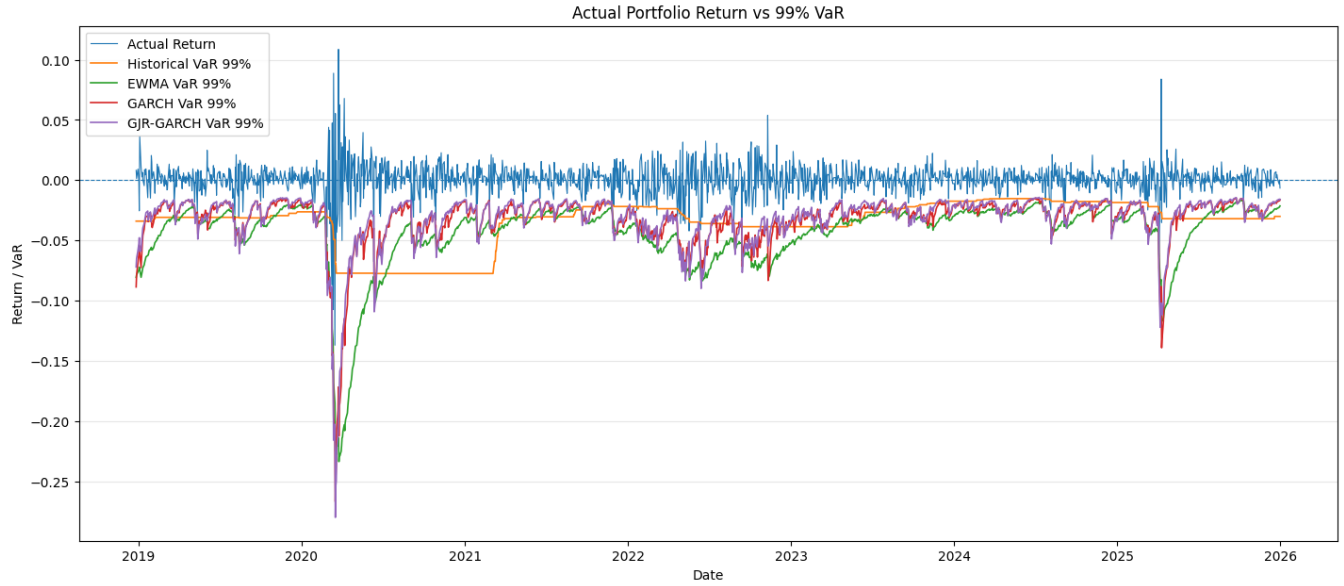


Exhibit 2a. Realized portfolio returns and the four aligned 99% VaR thresholds.

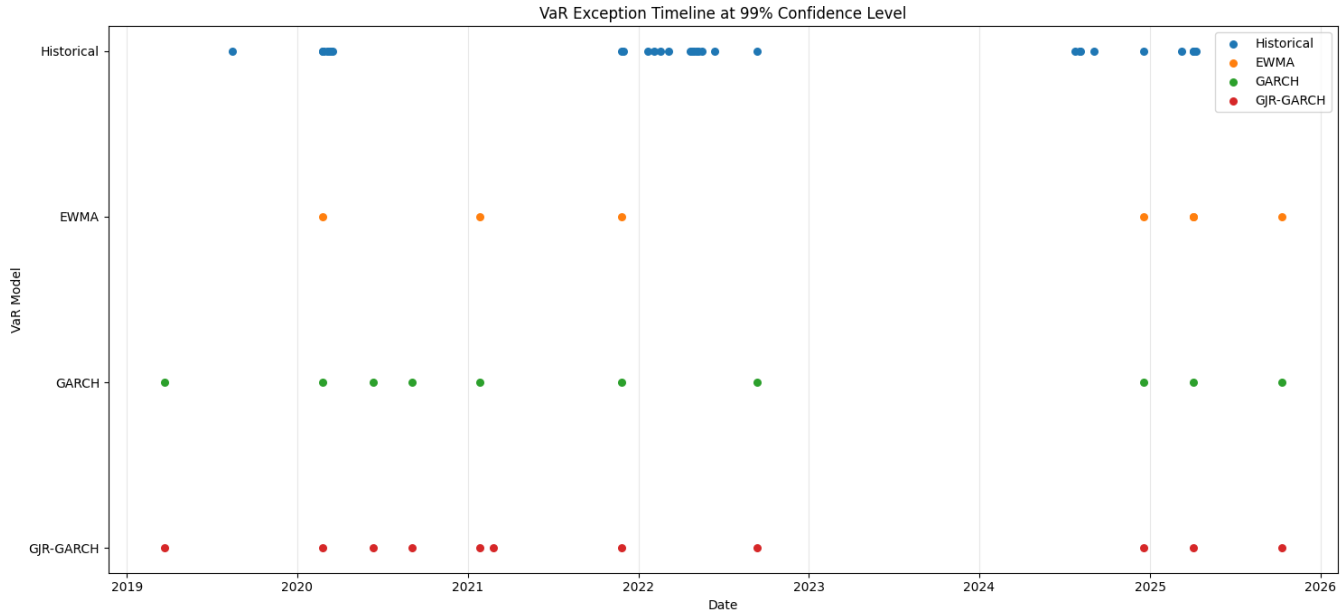


Exhibit 2b. Dates of the 99% exceptions across the four models during the holdout.

Reading the chart

A breach occurs when a realized return falls below a model's VaR threshold. The timeline shows when breaches cluster, while the return overlay shows how those events relate to the realized tail and forecast thresholds.

09 / RISK MAGNITUDE AND RANKING

Comparative results

The notebook ranks models first by the number of 99% backtesting tests passed, then by distance between the observed and target 1% exception rate. The ranking is descriptive of this sample and its chosen decision rule.

Rank	Model	Avg. 99% VaR	VaR SD	Avg. 99% ES	Exceptions	Tests passed
1	GJR-GARCH	-3.114%	2.227%	-4.299%	11	3 / 3
2	GARCH	-3.255%	2.209%	-4.343%	10	2 / 3
3	EWMA	-4.387%	2.855%	-6.458%	7	0 / 3
4	Historical	-3.456%	1.857%	-4.404%	32	0 / 3

Average forecast thresholds at 99%



A lower exception count is not automatically superior: EWMA records only 7 breaches but its observed 0.40% exception rate is far below the 1% target and all three tests reject calibration. GJR-GARCH balances a low threshold magnitude (-3.114% average cutoff) with 11 breaches and no test rejection. The ES values shown are model-based forecasts averaged across the out-of-sample period, as defined in the notebook.

Tail severity relative to the VaR threshold

Model	|ES| / |VaR| at 99%	Reading
Historical	1.27×	Tail average is about 27% deeper than the threshold.
EWMA	1.47×	Largest relative tail depth among the four models.
GARCH	1.33×	Tail depth is between Historical and EWMA.
GJR-GARCH	1.38×	Tail severity remains material despite the less negative average VaR.

Worst observed day

The worst realized portfolio return was -13.694% during the COVID period. It is common across models because it is an outcome in the shared realized return series, not a model forecast.

F3 / NOTEBOOK EXHIBIT

Model comparison: forecast paths and breach counts

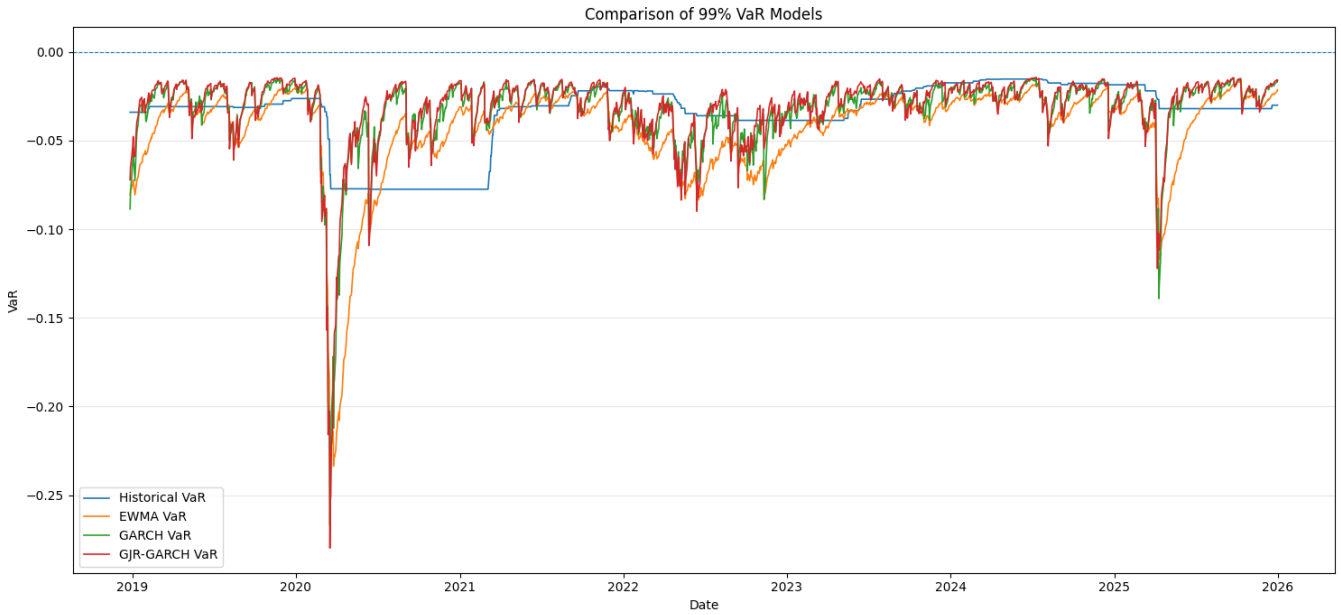


Exhibit 3a. 99% VaR forecast paths for Historical, EWMA, GARCH and GJR-GARCH.

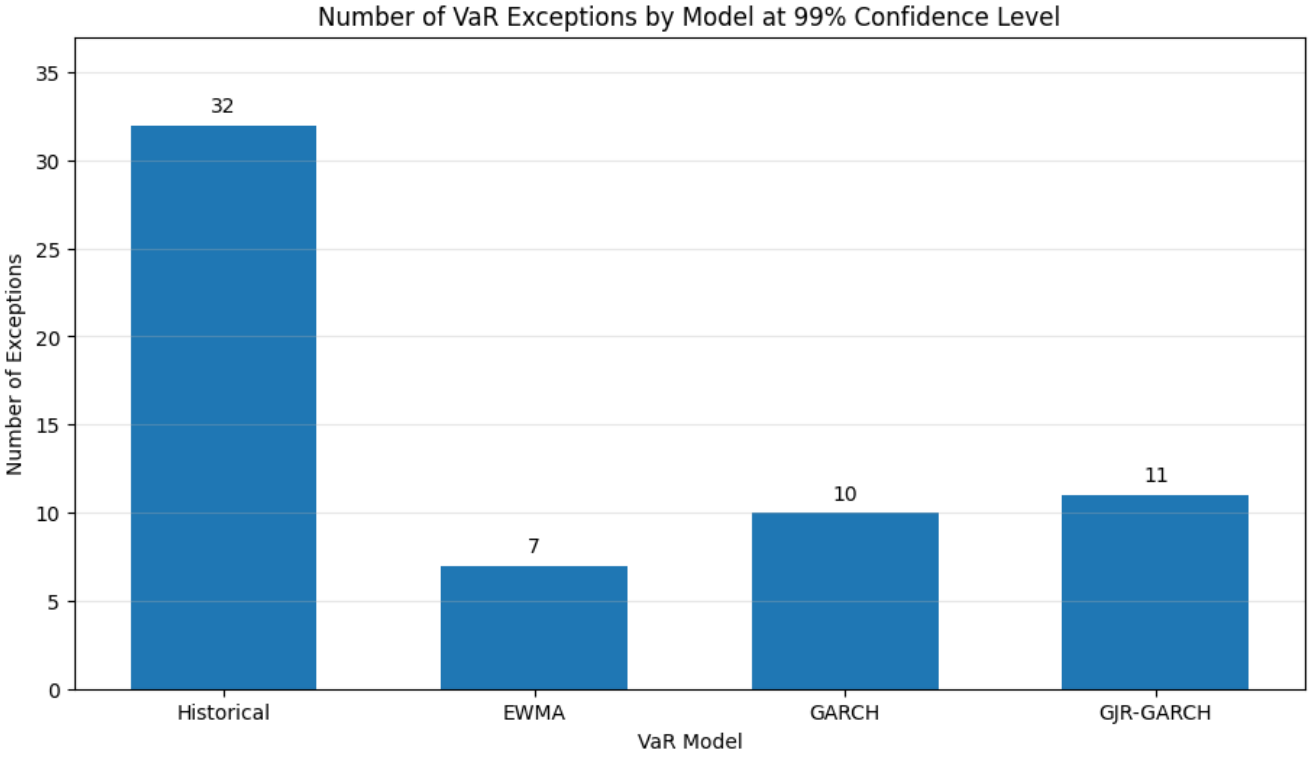


Exhibit 3b. Out-of-sample exception counts by model at 99% confidence.

Reading the chart

Historical VaR moves in steps as observations enter and leave its window, while volatility-scaled paths respond more continuously. The exception-count chart should be read with the coverage tests: fewer breaches can indicate conservative thresholds rather than better calibration.

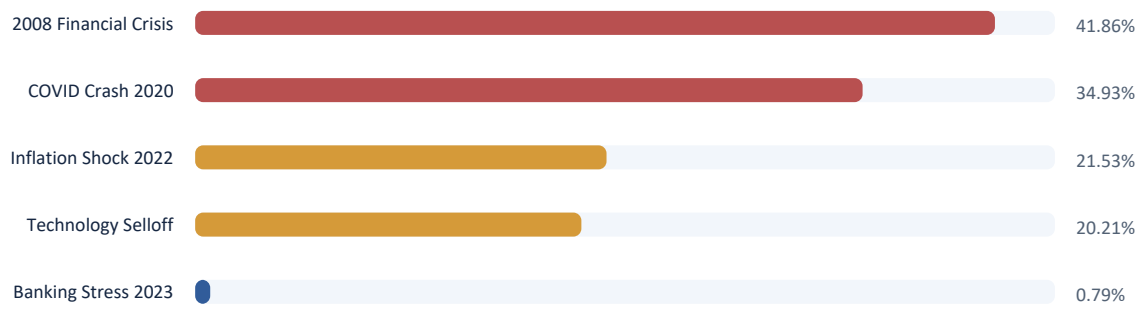
10 / CRISIS REPLAY

Historical stress episodes

Historical scenarios replay selected market windows using constituent returns available for each date range. Cumulative loss compounds over the episode. Scenario VaR, ES and worst-day loss summarize the daily return distribution inside that window. The different horizons mean cumulative loss and one-day tail measures should be read separately.

Episode	Window	Names	Cumulative loss	95% VaR	95% ES	Worst day
2008 Financial Crisis	15 Sep-20 Nov 2008	46	-41.86%	-7.11%	-8.44%	-9.14%
COVID Crash 2020	19 Feb-23 Mar 2020	50	-34.93%	-9.85%	-11.41%	-12.69%
Inflation Shock 2022	3 Jan-12 Oct 2022	50	-21.53%	-2.86%	-3.52%	-4.39%
Banking Stress 2023	8-20 Mar 2023	50	-0.79%	-1.63%	-1.90%	-1.90%
Technology Selloff	3 Jan-16 Jun 2022	50	-20.21%	-3.06%	-3.59%	-4.10%

Cumulative losses across episodes



The 2008 interval has the largest cumulative decline, but it is reconstructed from 46 available constituents rather than the full 50-stock universe. COVID has the largest 95% daily ES and worst daily loss among the listed historical episodes. The banking-stress window is shorter, which helps explain why its cumulative loss is modest relative to longer scenarios.

Scenario comparability

Historical scenario outputs depend on the exact start and end dates, available constituents, and portfolio reconstruction. They are controlled replays of selected episodes, not forecasts of future market paths.

11 / DESIGNED SHOCKS

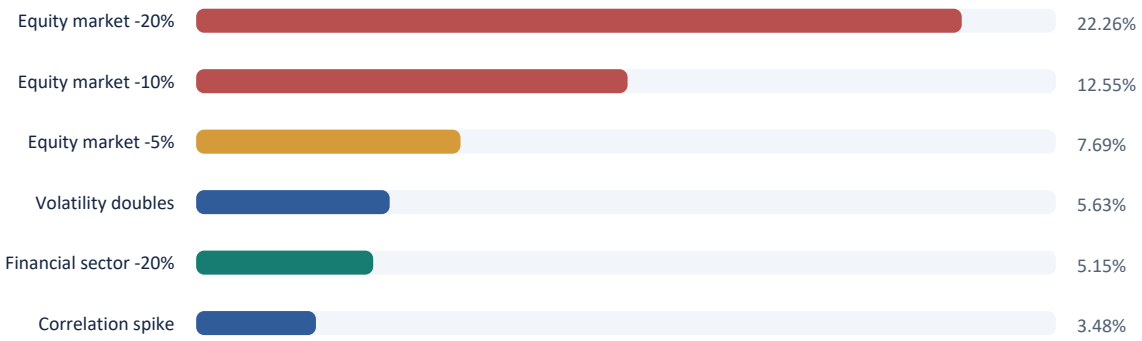
Hypothetical stress scenarios

Hypothetical scenarios impose shocks on the observed return or covariance structure. Their role is to expose sensitivity to specified assumptions. They are not calibrated forecasts, and a “worst-case” simulated result depends on the finite simulation design.

Scenario	Definition	Portfolio shock	95% VaR	95% ES	Worst day
Equity market -5%	Uniform broad-market decline	-5.00%	-6.58%	-7.69%	-17.05%
Equity market -10%	Uniform broad-market decline	-10.00%	-11.49%	-12.55%	-21.42%
Equity market -20%	Uniform broad-market decline	-20.00%	-21.33%	-22.26%	-30.15%
Volatility doubles	Scale volatility by 2×	-	-3.37%	-5.63%	-23.82%
Correlation spike	Converge correlations toward +1	-	-2.76%	-3.48%	-7.24%
Financial sector -20%	Shock 6 financial holdings	-2.40%[1]	-4.02%	-5.15%	-14.74%

[1] Direct portfolio impact for a uniform 20% shock to six of 50 equal-weight holdings. The cumulative-loss field is not defined for the volatility-doubling or correlation-spike calculations.

95% stressed ES by hypothetical scenario



The equity market -20% scenario has the most severe stressed ES in this set. Its stressed VaR and ES are tail measures of the shocked return distribution; the -20% portfolio shock is a separate imposed portfolio-level change. Under the sector scenario, six stocks receive a -20% shock, producing a direct first-order portfolio impact of -2.40% at equal weights.

The correlation scenario uses 50,000 Monte Carlo draws with random seed 42. The notebook checks the stressed covariance matrix's positive semidefiniteness before simulation. Reproducibility depends on the stated inputs and implementation.

Selected hypothetical tails against baseline

Scenario	95% VaR	95% ES	Change in interpretation
Baseline	-1.66%	-2.83%	Unstressed reference distribution
Volatility doubles	-3.37%	-5.63%	Tail magnitudes rise to about twice baseline
Correlation spike	-2.76%	-3.48%	Higher co-movement reduces diversification
Financial sector -20%	-4.02%	-5.15%	Sector shock affects six equal-weight names

F4 / NOTEBOOK EXHIBIT

Stress severity across VaR and Expected Shortfall

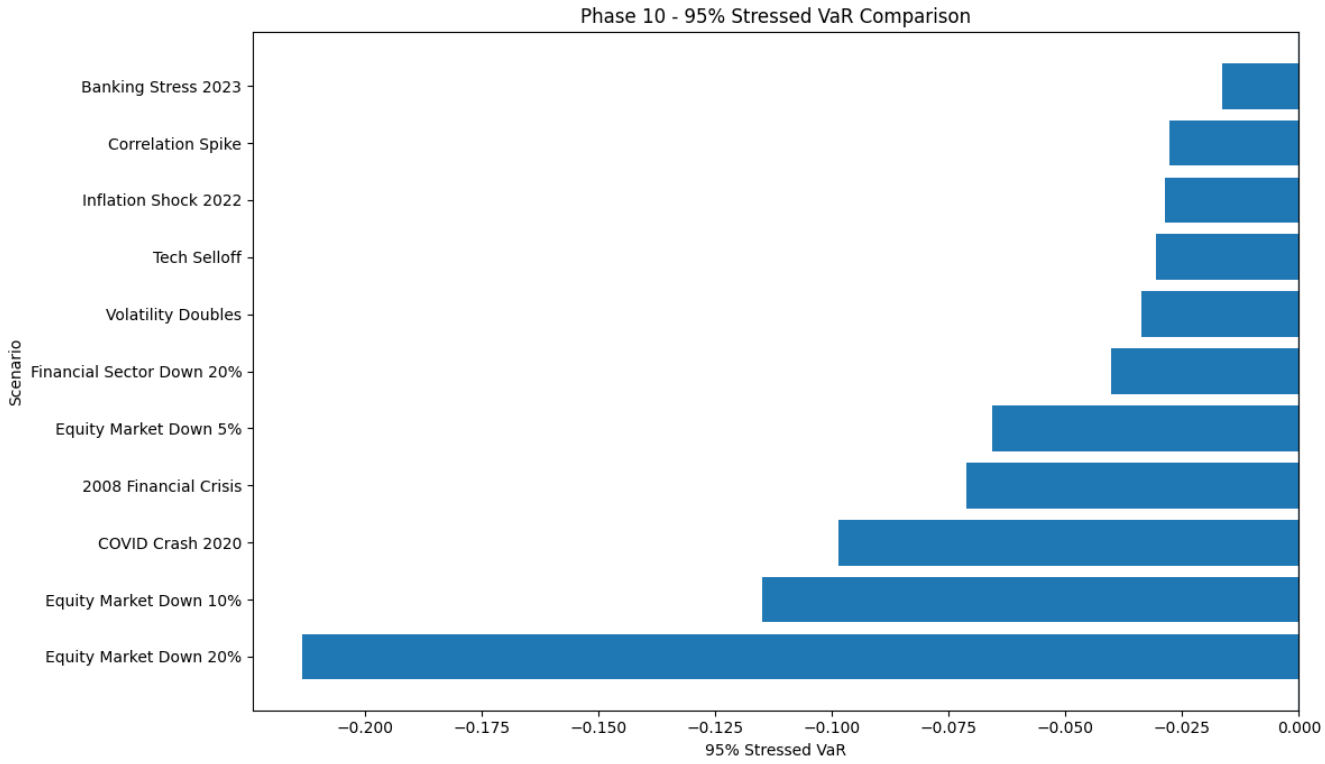


Exhibit 4a. 95% stressed VaR across the Phase 10 hypothetical scenarios.

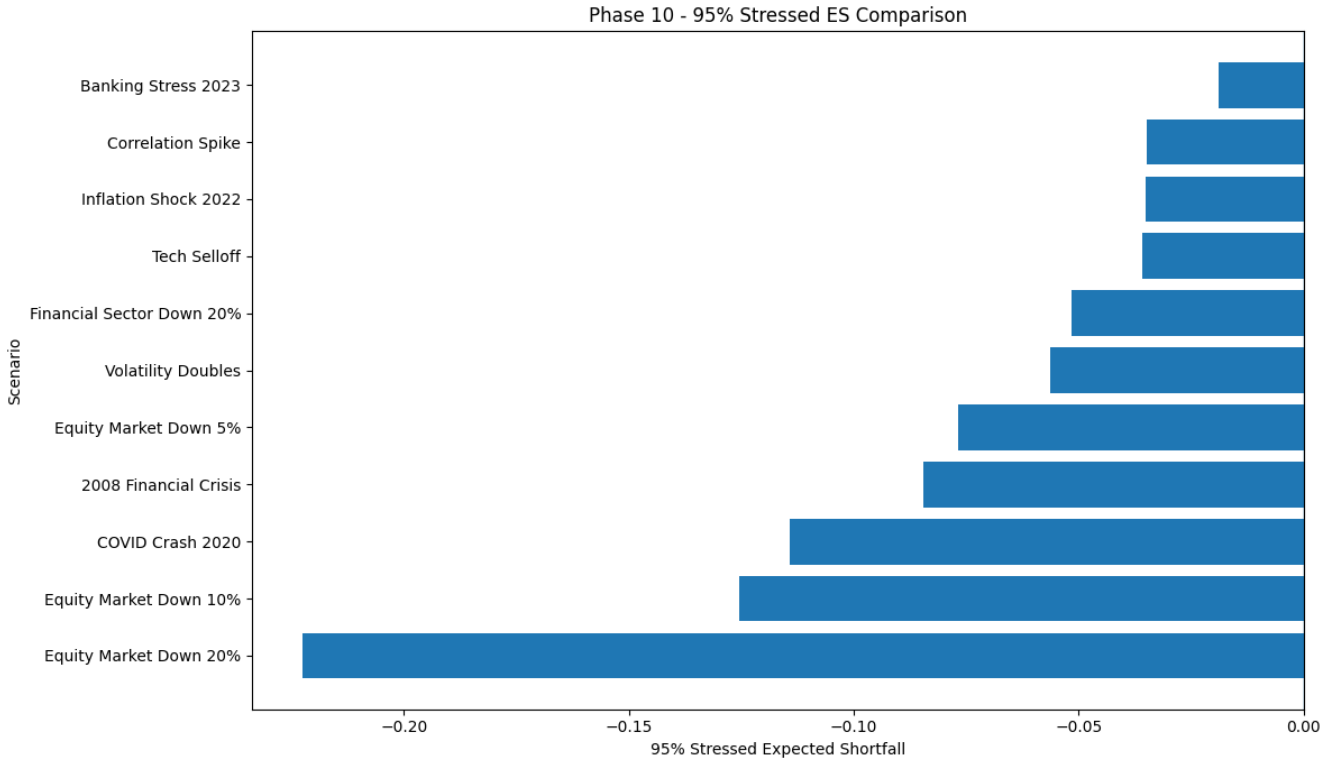


Exhibit 4b. 95% stressed Expected Shortfall across the same scenario set.

Reading the chart

Broad equity-market shocks produce the largest losses in both measures. VaR marks the tail threshold; ES averages losses beyond that threshold. These outputs are conditional on the imposed scenario definitions, not forecasts of their likelihood.

12 / REPRODUCIBILITY

Validation and limitations

The notebook runs explicit data, portfolio and VaR alignment checks. Its recorded summaries report 15 data/portfolio checks and 59 VaR/backtest checks passed. A separate recursive forecast consistency audit reports four passes for Historical, EWMA, GARCH and GJR-GARCH forecast construction. These checks support implementation consistency; they do not prove that model assumptions are true or that future returns will resemble the sample.

Audit layer	Notebook evidence	What it supports
Data and portfolio	15 checks passed	Date ordering/uniqueness, price/return integrity, weights and portfolio calculations.
VaR and backtest alignment	59 checks passed	Forecast availability, exception flags, threshold ordering and date alignment.
Recursive forecast consistency	4 checks passed	Forecast recursion matches the independently reconstructed equations.

Limitations

Issue	Implication for inference
Single equally weighted portfolio	Results may differ under alternative weights, holdings, rebalancing or constraints.
One evaluation period	The 2018-2025 holdout includes specific regimes; model ranking may change in other periods.
Normal GARCH innovations	Parametric tail assumptions may understate or misrepresent extreme outcomes.
Fixed initial estimation window	GARCH parameters are not re-estimated each day in the reported OOS design.
Historical stress reconstruction	2008 has 46 available names; scenarios use different windows and sample sizes.
No trading frictions	Turnover, transaction costs, liquidity and market impact are outside the analysis.
Exception-test power	With a finite sample, non-rejection is not proof of correct calibration.

How to read the ranking

The notebook's ranking is a rule-based summary of test outcomes at 99% plus distance from the 1% target. It is not an investment recommendation, a regulatory approval, or evidence that one model dominates in all market conditions.

Extensions that would strengthen the evidence

Extension	Research benefit
Walk-forward re-estimation	Shows whether parameter updates improve stability across regimes.
Alternative innovation distributions	Tests sensitivity of GARCH-family tail forecasts to heavy-tailed shocks.
Alternative weights and rebalance rules	Separates model effects from portfolio construction choices.
Repeated holdouts and longer histories	Provides more evidence on test power and ranking stability.
Liquidity and transaction-cost shocks	Adds implementation constraints to the risk interpretation.

13 / SYNTHESIS

Discussion and conclusion

Three conclusions follow from the notebook's results. First, the 99% backtests distinguish undercoverage from overconservatism: Historical Simulation produces too many exceptions, while EWMA produces substantially fewer than the nominal rate. Second, GARCH and GJR-GARCH produce exception rates below 1%, but only GJR-GARCH passes all three 99% tests under the notebook's decision rule. Third, stress scenarios reveal loss magnitudes that are not summarized by a single-day VaR threshold; the COVID and 2008 episodes show large cumulative drawdowns, while the equity-market -20% hypothetical shock produces the largest one-day ES in the reported scenario set.

The evidence is conditional on the portfolio, estimation design, sample and diagnostic tests. A stronger follow-up study would add rolling re-estimation, alternative innovation distributions, several portfolio weighting schemes, and repeated holdout or walk-forward comparisons. It would also preserve a separate scenario audit so that cumulative scenario losses, daily tail measures and model forecasts remain clearly distinguished.

Reproducibility note

All empirical results are transcribed from *VaR_Engine.ipynb*, including its out-of-sample backtest and validation outputs. The analysis focuses on return thresholds, volatility dynamics, exception tests and scenario losses.

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Appendix note: the dashboard is implemented in Streamlit and Plotly. The PDF is a static account of the notebook sample and should be regenerated if notebook inputs, dates, or model outputs change.